

Prediction and assessment of nanoprobe for tip-enhanced Raman spectroscopy: Data-driven artificial intelligence approach

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HIGHLIGHTS

- Explainable artificial intelligence (XAI) framework predicts nanoprobe geometry from electrochemical etching signals with a mean absolute percentage error of 9.47%.
- Perturbation-based XAI analysis indicates that the terminal portion of the etching current profile is a highly informative region for predicting the tip apex radius.
- Machine learning predictions are validated through finite-difference time-domain (FDTD) simulations and experimental TERS measurements on monolayer WS₂.
- An enhancement factor exceeding 20,000 is observed, supporting the accuracy and practical relevance of the AI-predicted tip performance.
- The proposed workflow enables pre-screening of high-performance TERS probes, reducing reliance on routine empirical testing and improving reproducibility.

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ABSTRACT

In Tip-enhanced Raman spectroscopy (TERS), the geometry of the metallic tip critically governs the strength of localized surface plasmon resonance and the achievable spatial resolution, yet reliable pre-experimental quality assessment remains challenging. Conventional electrochemical etching of gold nanoprobe produces substantial geometric variability, forcing researchers to rely on post-fabrication scanning electron microscopy (SEM) verification, a costly and labor-intensive bottleneck that limits high-throughput TERS experiments. Here, we introduce an integrated explainable artificial intelligence (XAI) framework that predicts tip geometry directly from real-time electrochemical etching current signals, achieving a mean absolute percentage error of 9.47 %. Through XAI perturbation analysis, we identify the final segment of the etching current trajectory as the most informative region for predicting nanoscale tip geometry. The predicted geometries are further evaluated through finite-difference time-domain (FDTD) simulations, and experimental validation is performed using scanning tunneling microscopy-based TERS measurements on monolayer WS₂. By enabling rapid screening of candidate probes, the proposed framework reduces the reliance on routine post-fabrication SEM characterization and shortens the nanoprobe development cycle, thereby improving experimental efficiency and enhancing the scalability of TERS probe development.

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1. Introduction

Tip-enhanced Raman spectroscopy (TERS) combines scanning probe microscopy (SPM) with Raman scattering to achieve sub-diffraction spatial resolution and highly localized signal enhancement. As a powerful tool in nanoscale spectroscopy, TERS enables probing of vibrational modes in low-dimensional materials with precision that surpasses conventional confocal techniques [1–6]. The key to TERS functionality strongly depends on the metallic nanoprobe, which simultaneously acts as a scanning probe and a localized plasmonic antenna. The apex radius and cone angle of the tip are critical parameters that determine both the spatial resolution and the electromagnetic enhancement factor, making precise control of tip geometry essential [7–9]. However, despite progress in automated electrochemical etching techniques that have improved the reproducibility of tip fabrication, evaluating tip quality before actual TERS measurements remains a practical challenge [10–12]. In most cases, tip performance is assessed only after experimental implementation, often relying on resource-intensive measurements or costly structural imaging such as scanning electron microscopy (SEM) [13]. These additional verification steps disrupt experimental workflows and introduce uncertainties that hinder reproducibility and slow down research progress. To address these limitations, recent studies have turned to machine learning (ML) as a transformative tool for materials characterization [14–18]. ML algorithms, particularly when integrated with explainable artificial intelligence (XAI), offer a data-driven approach to extract meaningful patterns and associations embedded within complex experimental data. XAI enhances this process by elucidating which features in the input data are most influential in the decision process of the ML model, providing both transparency and new scientific insights [19–22].

Accordingly, we present a data-driven ML framework that utilizes real-time current signals recorded during the electrochemical etching of gold nanotips. A Convolutional Neural Network (CNN) was trained to predict the apex radius and cone angle of the tips directly from the etching current profiles, without the need for structural imaging. Perturbation-based XAI analysis revealed that the final portion of the current signal is a decisive factor in determining tip geometry.

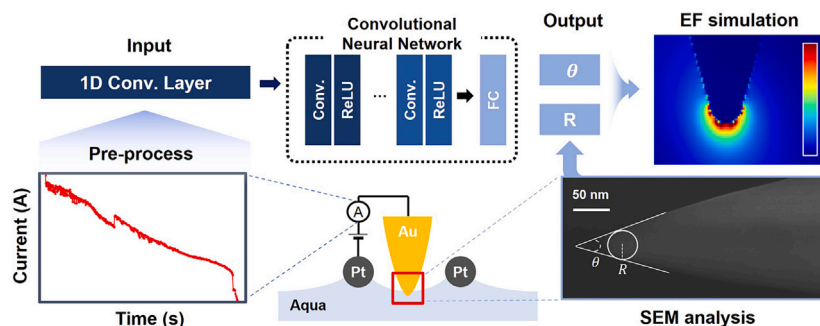
The accuracy of the model's predictions was validated through finite-difference time-domain (FDTD) simulations [13,23–25] and experimentally confirmed via TERS measurements conducted on monolayer WS₂. This framework offers a robust and interpretable solution for evaluating TERS tip quality prior to experimentation. It enables researchers to reduce reliance on empirical testing, improve reproducibility, and enhance experimental planning. In studies that rely on consistent signal enhancement and precise nanoscale characterization, this approach represents a practical and scientifically rigorous advancement for high-precision spectroscopy.

2. Results and discussion

2.1. Workflow of XAI-applied tip fabrication

Fig. 1 outlines the workflow of the proposed XAI model designed to predict the performance of TERS probes based on fabrication-derived signals. To construct a representative dataset for supervised learning, we used an electrochemical etching protocol in a KCl solution with a basic electrode arrangement, previously demonstrated to produce reproducible Au nanotips [11]. Real-time current signals were collected during the etching process, which subsequently served as the primary input features for the model. Because etching was performed under an applied square-wave voltage, the resulting current profiles reflected the dynamic progression of tip formation over time. However, significant variations in etching durations and morphological outcomes across diverse samples necessitated preprocessing to ensure standardization. Each current-time sequence was normalized and rescaled to a fixed 1 × 1024 dimension of a one-dimensional array, enabling uniform representation across samples. Geometrical labels were extracted by analyzing SEM images of the corresponding etched tips. The apex radius (R) and cone angle (θ) were manually annotated and assigned as the ground-truth outputs for supervised training. This process ensured a consistent pairing between each current profile and its associated physical geometry, as schematized in Fig. 1(a). To learn the correlation between current dynamics and tip shape, we adopted a 1-D CNN architecture, well-suited for extracting localized and hierarchical features from sequential data.

(a) Data / Training process



(b) Interpretation / verification

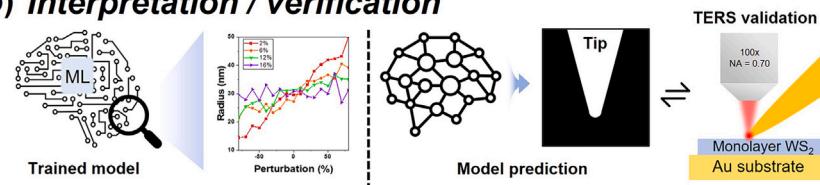


Fig. 1. Overview of the data acquisition, model training, and interpretation processes. (a) Overview of data acquisition and model training. Electrical current data obtained from electrochemical etching were pre-processed and used as input datasets for training a convolutional neural network (CNN). The apex radius (R) and cone angle (θ) of the fabricated tips, derived from SEM images, served as outputs for supervised training. (b) Model interpretation and experimental verification. Perturbation-based XAI techniques were applied to analyze the trained CNN, elucidating how input current profiles influence the predicted tip geometry. The predicted tip performance was subsequently verified through actual TERS measurements.

The dataset was randomly partitioned into training, validation, and test sets with an 8:1:1 ratio. Model training was performed on the training set, while the validation set was used to optimize hyperparameters such as batch size and epoch count.

Performance evaluation on the unseen test data revealed that discrepancies between training and test accuracy often reflected model overfitting (Figure S2). In such cases, iterative refinement of the network architecture and hyperparameter configuration was conducted until generalizable and interpretable performance was achieved. To interrogate the internal logic of the trained model, we implemented a perturbation-based interpretability approach. By selectively modifying localized regions within the input current profile and monitoring shifts in output predictions, we identified the specific signal segments most influential to the model's estimation of geometric parameters. This XAI-based analysis strategy is expanded upon in later sections.

2.2. Model training for performance prediction

In order to evaluate the practical applicability of our ML model, a comprehensive assessment was conducted by leveraging a test dataset that was not included in the training or validation process. This approach allowed us to probe the generalization ability of our ML model beyond its learned data distribution and to determine whether it could reliably infer geometric features from previously unseen etching signals.

Fig. 2 presents the predictive results of the model for three representative test samples. As shown in Fig. 2(a), each current profile was acquired under identical constant-voltage etching conditions and exhibits a broadly similar decay trend. Nevertheless, the actual geometry of the resulting tip cannot be directly inferred from these signals due to the nonlinear and complex nature of the etching dynamics. By direct observation, discerning meaningful differences in tip shape based solely on these current profiles would be extremely challenging [26, 27]. In contrast, the trained model successfully extracted subtle, high-dimensional patterns within the time-dependent current data, enabling it to predict distinct tip geometries, as illustrated in Fig. 2(b).

To validate these predictions, we performed SEM imaging for each tip (shown in Fig. 2(c)) to obtain the ground-truth values of apex radius and cone angle based on the actual measurement approach. Comparison

with predicted values revealed strong agreement, and the test set yielded a mean absolute percentage error (MAPE) of 9.47 %. This level of precision demonstrates our ML model's capacity to handle unfamiliar input signals and extract relevant geometric information from electrical data.

To further examine the plasmonic response associated with the predicted tip geometries in an application-relevant context, we simulated structure-dependent near-field enhancement using FDTD simulations implemented via the open-source Meep platform. This simplified configuration was used to isolate the influence of tip geometry on the plasmonic response. A TM-polarized plane wave ($\lambda = 632.8$ nm) was introduced under normal incidence, and the simulation was run with a spatial resolution of 1 nm. The dielectric function of gold was taken from the widely used Johnson and Christy dataset [28].

Despite the apparent similarity in their current inputs, the three predicted tips exhibited different local $|E|^2$ enhancement distributions in the FDTD simulations. These results, shown in Fig. 2(d), demonstrate that our ML model can recognize subtle but structurally performance-critical variations in the etching signals that translate into performance-relevant structural differences (e.g., tip radius, cone angle). This predictive capability indicates which tips are likely to exhibit favorable near-field enhancement characteristics prior to experimentation, offering a strategic advantage in TERS workflow design. Therefore, our model has the potential to not only improve experimental efficiency but also facilitate the targeted deployment of high-performance tips, suggesting that it could serve as a useful screening tool for enhancing both the accuracy and throughput of nanoscale spectroscopic analysis. However, the actual TERS performance ultimately depends on the full experimental configuration, including the tip-sample interaction in gap-mode geometries.

2.3. The XAI-applied model interpretation

To analyze the decision process of the trained ML model and thereby investigate which features most influence its predictions, we interpreted the ML model. Building on the physically interpretable framework developed in the previous sections, we interpreted our ML model to analyze which temporal regions of the etching current profile exert the greatest influence on the predicted nanoscale tip geometry.

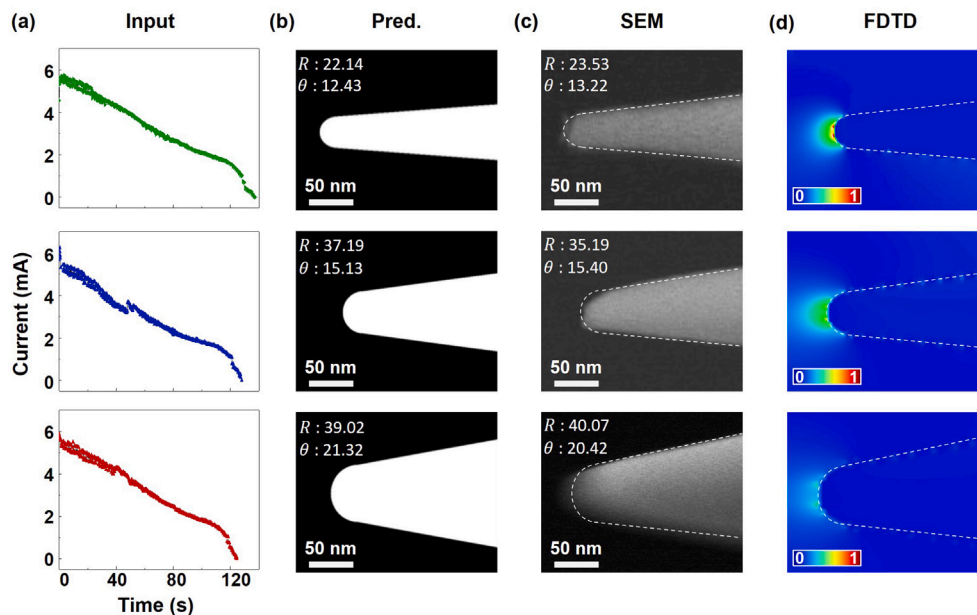


Fig. 2. Comparison between predicted and experimentally observed tip geometries. (a) Input current profiles acquired during electrochemical etching. (b) Predicted tip geometries visualized from R and θ estimated by the trained CNN model. (c) Corresponding SEM images of actual fabricated tips. The predicted tip shapes closely match the experimental results, validating the accuracy of the predictive model. Scale bars represent 50 nm.

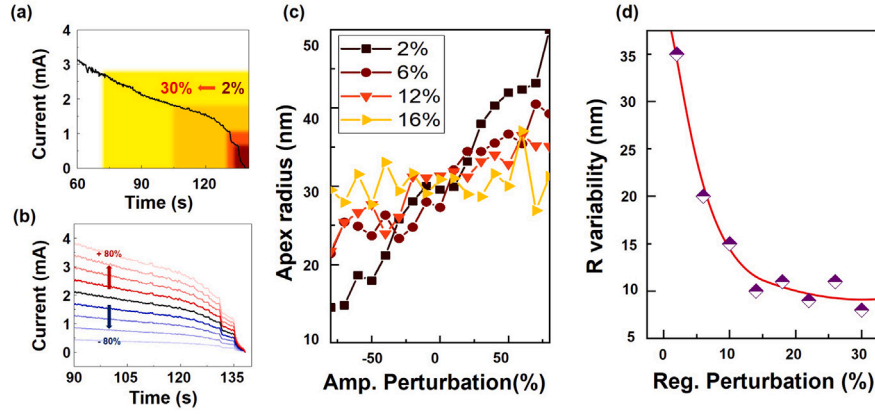


Fig. 3. XAI model interpretation through input perturbation analysis. (a) Perturbation of input current profiles was performed by varying the length of the altered data region from 2 % to 30 %, starting from the endpoint of the etching current signal. (b) Input current profiles were perturbed by fixing the length of the data region and varying the magnitude of perturbation from −80 % to +80 %. (c) Changes in the predicted R in response to input perturbation. (d) Corresponding changes in the predicted θ .

To systematically evaluate input sensitivity, we implemented a perturbation-based analysis strategy. Two schemes were designed for this purpose. As shown in Fig. 3(a), we progressively extended the perturbed region of the current signal from 2 % to 30 % of the signal length, with increments of 4 %, starting from the signal endpoint. This design reflects the hypothesis that the final stages of etching play a critical role in shaping the tip geometry. In the second scheme (Fig. 3b), we fixed the length of the perturbed region and varied the perturbation magnitude from −80 % to +80 % to examine the model's response to localized distortions in the input.

Within the selected region, the perturbation was applied through multiplicative scaling of the current signal accordingly.

$$I'(t) = I(t)(1 + \alpha)$$

where $I(t)$ denotes the original current signal, $I'(t)$ is the perturbed signal, and α represents the perturbation magnitude varied between −0.8 and +0.8. The results of these perturbations, shown in Fig. 3(c), reveal that even when the perturbation magnitude is held constant, the location of the modified region significantly affects the predicted apex radius. The most substantial changes occur when perturbations are confined to the terminal segment of the signal, suggesting that this portion contains especially influential information. These findings imply that although the overall current profile is relevant, this portion contains particularly informative features for predicting the final tip geometry. Fig. 3(d) further illustrates that variations in the final segment of the etching current profile are strongly associated with changes in the predicted tip morphology, indicating the model assigns substantial weight to this region. As the perturbed region moves farther from the signal endpoint, the variability in the predicted radius steadily declines and eventually converges. This trend indicates that the model, although trained on the full current profile, has autonomously learned to assign greater importance to the terminal segment without the need for explicit labeling. In contrast, the predicted cone angle (θ) showed relative insensitivity to similar perturbations. As summarized in Figure S4, this stability suggests that θ is more strongly influenced by global fabrication conditions, including variables such as electrolyte concentration and total etching time, rather than by localized variations in the etching current profile.

Taken together, these results confirm that our XAI framework not only provides model transparency but also validates the physical relevance of temporally localized input features and offers a deeper understanding of how current dynamics translate into geometric outcomes. This interpretability can serve as a practical guide for further refinement of the etching process and model-driven experimental design.

2.4. Experimental verification

To experimentally validate the performance predicted by our trained model, we performed a TERS measurement using a tip fabricated through electrochemical etching. Fig. 4(a) shows the etching current profile, which was preprocessed as described earlier and served as the model input. The predicted geometric parameters, including the apex radius and cone angle, were subsequently applied as input conditions for finite-difference time-domain (FDTD) simulations. As shown in Fig. 4(b), the simulation configuration involved a gold tip inclined at 60° relative to the horizontal plane, positioned above an Au substrate to replicate the experimental arrangement. The simulation results indicated an electric field enhancement ($|E|^2$) of approximately 2×10^4 , demonstrating strong near-field localization consistent with the predicted tip geometry.

We then conducted TERS experiments to evaluate the optical enhancement capability of the same tip. An STM-based TERS setup was employed, where a tunneling current was used to approach a monolayer WS_2 sample placed on a conductive Au substrate. Figure 4(c) presents Raman spectra obtained under confocal and TERS conditions. Upon tip illumination, both signal intensity and spectral features were significantly enhanced, primarily due to localized surface plasmon resonance (LSPR) at the tip apex [29–31].

To quantify the enhancement, we calculated the contrast of the A_{Ig} peak between the TERS and confocal spectra using:

$$\text{Contrast} = \frac{I_{\text{TERS}} - I_{\text{Confocal}}}{I_{\text{Confocal}}}$$

This contrast value, in conjunction with the predicted tip geometry, was used to estimate the TERS enhancement factor (EF) via:

$$EF = \text{Contrast} \left(\frac{R_{\text{laser}}}{R_{\text{tip}}} \right)^2$$

where R_{laser} denotes the effective radius of the diffraction-limited laser spot and R_{tip} represents the effective near-field interaction radius at the tip apex. The laser spot radius was estimated using the standard approximation for a focused beam,

$$R_{\text{laser}} \approx \frac{0.61\lambda}{NA}$$

where $\lambda = 632.8 \text{ nm}$ and $NA = 0.7$, yielding $R_{\text{laser}} \approx 550 \text{ nm}$. The effective tip radius used in the EF calculation was taken from the ML-predicted apex radius ($R_{\text{tip}} = 27.2 \text{ nm}$). The corresponding SEM image of the experimentally fabricated tip and the measured apex radius are provided

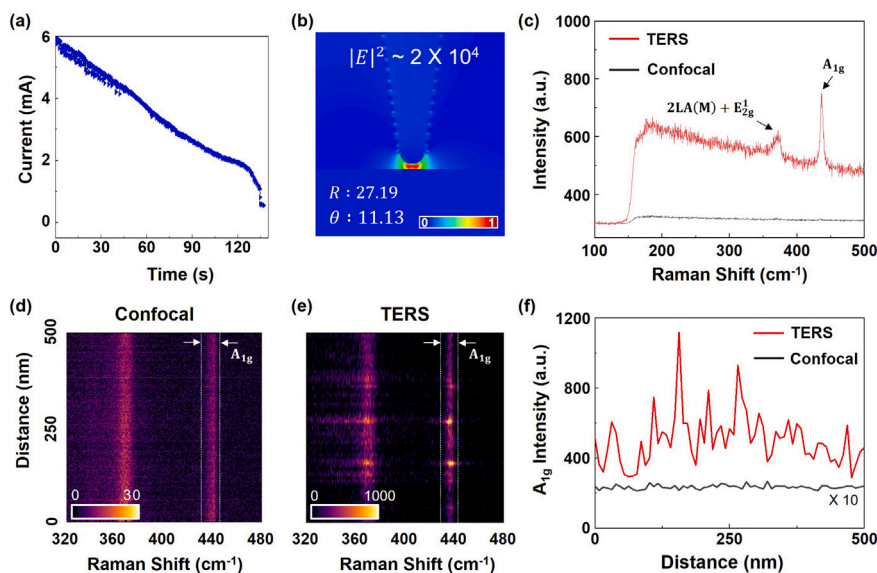


Fig. 4. Experimental validation of predicted tip performance through TERS measurements. (a) Electrochemical etching current profile used as input data for predicting tip geometry. (b) Predicted electric field intensity ($|E|^2$) distribution obtained by FDTD simulation based on the predicted tip apex radius ($R = 27.19$ nm) and cone angle ($\theta = 11.13^\circ$). (c) Comparison of Raman spectrum measured with tip (Tip in, TERS) and without tip (Tip out, confocal). (d, e) Raman line scan images of monolayer WS_2 obtained by confocal and TERS, respectively. (f) Corresponding intensity profiles of the WS_2 A_{1g} mode extracted from (d), (e), clearly demonstrating the enhanced spatial resolution and signal intensity in TERS measurements.

in the Supplementary Information (Figure S7), showing good agreement with the predicted value.

Assuming $R_{\text{laser}} \approx 550$ nm and $R_{\text{tip}} = 27.2$ nm, the enhancement factor was estimated to be greater than 2×10^4 , which is consistent with typical values reported for gap-mode TERS measurements.

Assuming a laser spot radius of $R_{\text{laser}} \approx 550$ nm and a tip apex radius of $R_{\text{tip}} = 27.2$ nm, the calculated EF was approximately greater than 20,000 [10,13,23,32–34]. This result closely aligns with the field intensity predicted by the FDTD simulation (Fig. 4b), supporting the accuracy and practical utility of the model.

In addition to single-point spectra, a line scan of Raman scattering was performed across a 500 nm region with 7.8 nm step intervals. The confocal scan (Fig. 4d) exhibited negligible spatial variation. In contrast, the TERS map (Fig. 4e) revealed spatially dynamic changes in both A_{1g} and E_{2g}^1 Raman modes, including observable frequency shifts. These results indicate strong near-field confinement and highly localized probe-sample interactions, characteristic of TERS. The spectral variations observed along the line scan are likely associated with local heterogeneity within the CVD-grown WS_2 flake, such as nanoscale variations in strain, defect density, or the local electronic environment. Because the TERS signal originates from the strongly confined near-field at the tip apex, the measured spectra are highly sensitive to such nanoscale variations in the local sample environment. The corresponding A_{1g} mode intensity profile, extracted from the same dataset, is shown in Fig. 4(f). This profile exhibits sharp, spatially confined peaks that were absent in the confocal case, thereby confirming that the predicted tip geometry enables robust near-field enhancement and facilitates high-resolution chemical analysis at the nanoscale. Additional validation experiments using multiple probes with different apex radii are provided in Figure S6. The measurements indicate that the spatial transition becomes progressively broader for probes with larger apex radii, consistent with the commonly observed dependence of TERS spatial resolution on probe apex geometry.

3. Conclusions

In this work, we introduced an eXplainable AI framework capable of predicting the geometry and optical response associated with the

predicted tip geometries of TERS probes directly from electrochemical etching current signals. The model, trained on time-resolved current profiles using a CNN, achieved accurate estimation of apex radius and cone angle with a MAPE of 9.47 %. Interpretation through perturbation-based XAI revealed that the terminal portion of the etching signal plays a dominant role in shaping the tip apex, providing mechanistic insight into the formation process. The model's predictions were further validated through FDTD simulations and STM-based TERS measurements performed on monolayer WS_2 . The enhancement factor calculated from the predicted tip geometry showed consistency with both simulated near-field distributions and experimentally observed Raman intensities. In addition, spatially resolved TERS line scans revealed local spectral features that are not accessible by confocal techniques, thereby verifying the effectiveness of the predicted tip in practical applications.

Altogether, the proposed workflow, which combines machine learning, physical modeling, and experimental validation, offers a robust methodology for pre-screening nanoprobe. This strategy has the potential to reduce the need for iterative experimental testing and may support the targeted deployment of high-performance tips, accelerating research in nanoscale vibrational spectroscopy. Beyond the present framework, future studies may extend the model to directly estimate TERS performance metrics, such as spatial resolution or enhancement factor, enabling more predictive screening of probes tailored to specific experimental requirements.

4. Methods

4.1. Au tip fabrication

The Au nanotips used throughout this study, including those used for dataset construction and TERS measurements, were fabricated via an electrochemical etching process using optimized conditions based on previously reported protocols [13]. The etchant solution was prepared by dissolving KCl powder in deionized (DI) water to obtain a concentration of 2.8 mol/L under ambient conditions. The solution was mixed using a vortex mixer until the KCl powder was completely dissolved. Subsequently, the KCl solution was combined with 99.9 % anhydrous ethanol at a volume ratio of KCl solution:ethanol = 4:1. Because the addition of ethanol induced partial precipitation, the mixture was

further agitated using the vortex mixer to ensure homogeneous dispersion. Finally, the effective concentration of KCl in the etchant was adjusted by adding 1 mL of DI water to 19 mL of the KCl-ethanol mixture.

Au nanotips were fabricated using a homemade electrochemical etching setup. An Au wire (0.1 mm diameter, 99.95 %, Nilaco, Japan) served as the anode, while a Pt ring electrode (0.2 mm diameter, 99.98 %, Nilaco, Japan) acted as the cathode. The Au wire was immersed into 20 mL of the etchant to a depth of approximately 0.1 mm. A square-wave voltage waveform was applied between the electrodes using a waveform generator, with electrical parameters of 2.417 V (maximum voltage), −250 mV (minimum voltage), a frequency of 300 Hz, and a duty cycle of 77 %. The etching process proceeded in a self-terminating manner until the lower part of the wire detached, producing a sharp Au nanotip.

4.2. Tip geometry characterization

For SEM characterization, we used a field-emission scanning electron microscope (S-4800, Hitachi) operated at an accelerating voltage of 10 kV and a fixed magnification of 30,000 $\times$. The working distance was set to approximately 12 mm, and all measurements were performed under high-vacuum conditions. The system provides a spatial resolution of 1.4 nm at 10 kV, which was sufficient for resolving the fine geometrical features of the fabricated tips.

4.3. Model training

The proposed neural network architecture processes electrochemical etching current signals by first normalizing the raw inputs and reshaping them into a tensor of shape (batch size, 1, 1024). The model predicts two target parameters, namely the cone angle and apex radius, from the normalized current profiles. A one-dimensional convolutional feature extractor is employed, consisting of three sequential Conv1D layers with 16, 32, and 64 filters, respectively (kernel size = 3, padding = 1). Each convolutional layer is followed by a ReLU activation function [35] and a max-pooling layer with a pool size of 2, progressively reducing the sequence length from 1024 to 128. The resulting feature representation, with a shape of (batch size, 64, 128), is flattened into an 8192-dimensional vector. This vector is passed through a fully connected layer with 64 units and subsequently mapped to the two output parameters using a final linear layer. The dataset comprised 200 experimentally collected samples, each labeled with the corresponding geometric parameters (apex radius and cone angle). Model training and evaluation were performed by splitting the dataset into training, validation, and test subsets with an 8:1:1 ratio. A fixed random seed (seed = 42) was used for dataset partitioning to ensure reproducibility. All preprocessing, model training, and interpretation analyses were implemented in Python using the PyTorch [36] framework. The training protocol employed an initial learning rate of 0.01, a batch size of 32, and 75 training epochs. The Adam optimizer and mean squared error (MSE) loss function were used throughout training. Model convergence and potential overfitting were monitored by comparing training and validation losses (Supplementary Information, Figures S1 and S2). All experiments, including model training, inference, and interpretation analyses, were conducted using a single NVIDIA RTX 3090 Ti GPU.

4.4. Sample preparation

Monolayer 2H-WS₂ flakes were synthesized using a chemical vapor deposition (CVD) method. SiO₂/Si substrate (2 cm $\times$ 2 cm) was used as the growth platform. Prior to deposition, the substrate was cleaned by sequential sonication in acetone and isopropyl alcohol (IPA) for 10 minutes each.

For precursor preparation, 0.1 mL of hydrazine hydrate (80 % aqueous solution, Sigma-Aldrich) was mixed with 10 mL of 0.01 M Na₂WO₄·2H₂O (99 %, Sigma-Aldrich). The resulting solution was dropped onto the center of the cleaned substrate and spin-coated at 3000 rpm for

1 minute. Sulfur powder (99.98 %, Sigma-Aldrich) was placed upstream in the low-temperature zone, while the substrate was positioned downstream in the high-temperature zone of a quartz tube furnace. Before initiating growth, the system was purged with argon gas at a flow rate of 600 sccm for 20 minutes at room temperature.

During growth, the sulfur and substrate zones were ramped to 210°C and 835°C, respectively, within 7 minutes under a mixed gas flow of 550 sccm Ar and 8 sccm H₂. Sulfur vapor was transported downstream to the substrate via the carrier gas, and the growth temperature was maintained for 9 minutes. After growth, the system was allowed to cool naturally to room temperature by opening the furnace.

The as-grown WS₂ flakes were subsequently transferred onto a gold substrate using a PMMA-assisted wet transfer process. A PMMA layer was first spin-coated onto the WS₂/SiO₂/Si substrate to support the flakes during transfer. The underlying SiO₂ layer was then etched using a hydrofluoric acid (HF) solution, releasing the PMMA/WS₂ film from the substrate. The floating film was transferred onto the Au substrate and dried under ambient conditions. Finally, the PMMA layer was removed by sequential rinsing in acetone, isopropyl alcohol (IPA), and deionized (DI) water.

4.5. TERS measurements

The TERS system (NTEGRA Spectra, NT-MDT) integrates a confocal Raman scattering setup with scanning tunneling microscopy (STM), where the STM functionality was employed exclusively for approach control. The tunneling current was set to approximately 0.2 nA with a bias voltage of 0.1 V under ambient conditions to ensure stable tip-sample contact. To minimize lateral drift during Raman mapping and line-scan measurements, the STM system was operated in closed-loop scanning mode. TERS measurements were performed using a 632.8 nm excitation laser and an objective lens with a numerical aperture (NA) of 0.7 (Mitutoyo). Multispectral Raman spectra were acquired using a spectrometer equipped with an 1800 grooves/mm grating (blazed at 500 nm) and a CCD detector (Andor) cooled to −60°C. All measurements were conducted using the same electrochemically etched gold tip.

4.6. Finite-difference time-domain simulations

Optical responses under gap-mode TERS conditions were investigated using FDTD simulations implemented in MEEP (PyMeep v1.29.0).

The simulation geometry was implemented in two dimensions and consisted of a metallic tip positioned above a metallic substrate to represent the tip-sample junction. The tip geometry was modeled as a conical body with a rounded apex, consistent with the experimentally observed probe morphology. The tip-substrate gap distance was set to 5 nm to reproduce the gap-mode TERS configuration used in the experiments.

Illumination was modeled using a broadband Gaussian-modulated plane wave (line source) injected from one side of the computational domain. Oblique illumination was implemented by applying a Bloch phase shift to the source, enabling the representation of the side-illumination configuration used in the experiments. To reproduce the experimental TERS configuration, the illumination angle was set to 60° with respect to the substrate surface in the gap-mode simulations.

The computational domain was defined as a rectangular region of 3 μ m $\times$ 0.1 μ m in the horizontal and vertical directions, respectively. Perfectly matched layers (PMLs) were applied at the domain boundaries to absorb outgoing electromagnetic waves. A spatial grid resolution of 1 nm was used throughout the simulation to accurately resolve the near-field distribution in the tip-sample gap region.

To evaluate the optical response, two types of monitors were employed. Flux monitors were placed at reflection and transmission positions to calculate spectral quantities such as extinction. In addition, point monitors located at the tip-sample gap were used to record the local electric-field components. Full-field monitors at selected wavelengths were evaluated using discrete Fourier transform (DFT) fields.

The local near-field enhancement was quantified using the electric-field intensity enhancement factor defined as $|E|^2/|E_0|^2$, where E represents the local electric field and E_0 denotes the incident field amplitude. For comparison across different tip geometries, the calculated quantities were normalized to the corresponding empty reference simulation.

CRedit authorship contribution statement

Hyung Chan Suh: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Taeheon Kim:** Methodology, Formal analysis, Data curation, Conceptualization. **Dong-Joon Yi:** Methodology, Formal analysis. **Dong Hyeon Kim:** Data curation. **Sung Hyuk Kim:** Data curation. **Jaekak Yoo:** Validation, Methodology. **Seungho Bang:** Writing – review & editing, Validation. **Dohyeon Lee:** Data curation. **Ji Hong Kim:** Methodology. **Yo Seob Won:** Validation, Resources. **Ki Kang Kim:** Writing – review & editing, Validation, Resources. **Mun Seok Jeong:** Writing – review & editing, Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data for this article can be found online at doi:10.1016/j.matdes.2026.116066.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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